BIBI ZAINAB

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ECE GRADUATE | PCB DESIGN | KiCad | EMBEDDED SYSTEMS

PROFESSIONAL SUMMARY

Electronics and Communication Engineering graduate with project experience in embedded systems, sensor interfacing and Arduino-based prototyping, and training in PCB design using KiCad (schematic capture and PCB layout fundamentals). Built a working reverse vending machine with sensors, servos, GSM and Bluetooth, and also has Python, machine-learning and cybersecurity project experience. Seeking an entry-level role in PCB design, embedded systems or electronics hardware.

TECHNICAL SKILLS

- **PCB Design:** KiCad, schematic capture, PCB layout fundamentals
- **Embedded Systems:** Arduino Mega, Embedded C/C++, sensor and actuator interfacing (load cell with HX711, inductive sensor, servo motor, I2C LCD, keypad), UART modules (GSM, Bluetooth HC-05), EEPROM
- **VLSI Fundamentals:** Verilog HDL basics, RTL concepts, introductory FPGA and ASIC concepts
- **Programming & Tools:** Python, TensorFlow/Keras, Scikit-learn, Flask, Git/GitHub, Linux, Arduino IDE
- **IT & Security:** Kali Linux, VMware, Windows Server, Active Directory, AWS EC2, Apache, Nginx

PROJECTS

Reverse Vending Machine (Team project, 4 members) | github.com/zk946190-ui/reverse-vending-machine
Arduino Mega, Embedded C/C++, HX711 load cell, inductive sensor, GSM 4G module, HC-05 Bluetooth, 20x4 I2C LCD

- Built a campus recycling machine that sorts plastic, metal and glass bottles: inductive sensor for metal, load cell (HX711) to separate plastic from glass, two servo motors to drop and route each bottle.
- Implemented keypad USN login, reward points stored in EEPROM, a Bluetooth admin panel (ADD USER / DELETE USER / LIST) and GSM SMS notifications after each deposit and when the bins are full.
- Tested with 50 bottles per type: 90% (plastic), 100% (metal) and 80% (glass) sorted correctly.

Course Recommendation Agent | github.com/zk946190-ui/AI-agent
Python, Groq LLM API, JSON

- Built an AI agent that reads student profiles and a course catalogue (JSON) and uses the Groq LLM to generate a personalised learning path, prerequisites, timeline and career advice.

AI-Based Phishing URL Detection | github.com/zk946190-ui/phishing-url-detection
Python, Scikit-learn, Pandas, TF-IDF, Random Forest, Flask

- Trained a TF-IDF + Random Forest classifier to label URLs as phishing or safe (78.1% accuracy on 498 test URLs) and built a Flask web dashboard for real-time checks.

Skin Cancer Detection using CNN | github.com/zk946190-ui/skin-cancer-detection-cnn
Python, TensorFlow/Keras, EfficientNetB7

- Classified skin lesion images as benign or malignant; improved test accuracy from 45.5% (CNN from scratch) to 83.3% using EfficientNetB7 transfer learning.

INTERNSHIP

Cybersecurity Intern — GTTC, Hubli | February 2026 – May 2026

- Studied networking fundamentals (IP addressing, subnetting, ports, protocols, OSI model) and set up Kali Linux, Windows Server and Ubuntu in VMware.
- Managed users, groups and organizational units in Active Directory; launched AWS EC2 servers and configured Apache and Nginx (hosting, reverse proxy, load balancing).
- Practised PowerShell and Bash scripting; completed a capstone project, the phishing URL detector above.

EDUCATION

B.E. in Electronics and Communication Engineering | 2022–2026
Government Engineering College, Talakal, Koppal (Visvesvaraya Technological University, Belagavi) | CGPA: 7.95/10

TRAINING & CERTIFICATIONS

- PCB Design Training — KiCad (schematic capture and PCB layout fundamentals)
- Cybersecurity and Ethical Hacking — Cisco learning certificates
- Cybersecurity internship completion certificate — GTTC, Hubli